Reflow-Oven Controller

User Manual

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1 Introduction

The purpose of this document is to explain how to use the reflow oven.

This device is designed to convert a standard consumer oven into a PCB reflow soldering station. The main objective is to precisely control the oven's heating element to follow a specific temperature profile required for soldering surface-mount components. The controller manages the AC mains power to the heater, monitors the internal temperature in real-time, and provides a user interface for operation and status monitoring.

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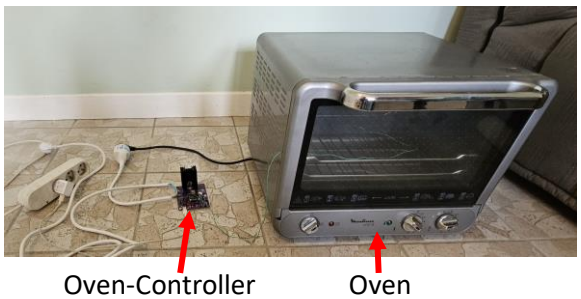
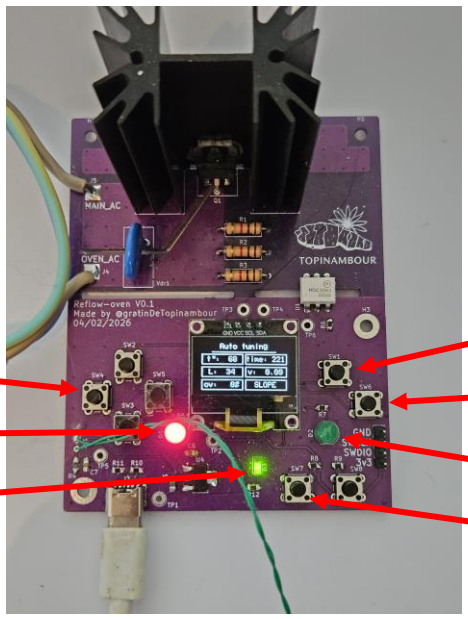
2 System Overview

2.1 Functional Description

The Reflow Oven Controller operates with a regular mini oven.





Oven Controller: the Board that control the Oven

Oven: Any Oven that can go up to 240°C under 2400W (for 220V 50Hz main supplies only). You need a simple furnace without power-loss security (because the controller will switch on and off the main AC if the oven has a power loss safety the controller will not be able to turn it back on)

Cross button: Button used for navigation.

Oven On LED: This Red LED will turn on each time the Oven is switched on.

Power On LED: Green LED light on if the board is correctly powered through the 5V USB cable.

Return Button: Return to the previous menu.

Valid Button: Use this button to go to the next menu.

Green LED: switch on and off each second.

Reset Button: Reset the board.

2.2 Power on

Cut the red wire of any electrical extension cord and solder each side of the cut wire to Main AC and Oven AC on the controller board.

Check that the BTA16 is correctly soldered to the board, and the heat sink is correctly placed with **thermal paste.**

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Connect a 5v USB cable to the board. The Power-on LED should light up, and the Green led should blink. If the screen is not turned on, press Reset.

Plug the Oven into the extension cord.

After this step the board will be connected to the main AC. **DO NOT TOUCH ANY PART on the top side of the controller board.**

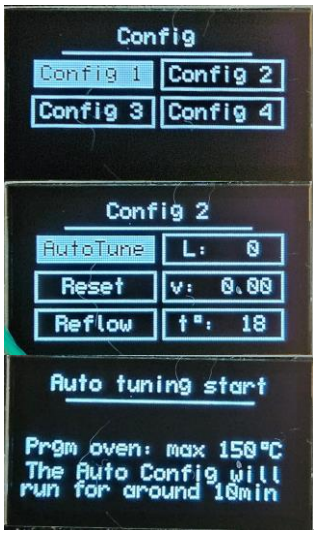
Plug the extension board on main AC.

3 How to use

Once the display is on, press the valid button. You will be able to choose between 4 different configurations if you have multiple ovens or if you want to try some modifications. When using a new configuration all parameters are set to zero, you need to setup those values either by hand or with the autotune method, or else the controller will not be able to control the oven.

3.1 Autotune

The autotune method takes around 10min. The controller will record the speed of the oven (v) the reaction time (L) and the overshoot (ov).



-chose a configuration and press valid

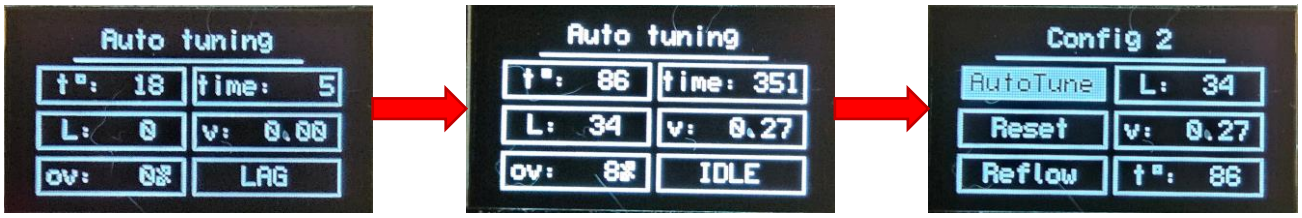
-select autotune and press valid

-before starting the autotune you need to setup the oven

-place the sensor in the middle of the oven

-set the oven to max 150°C and start the oven (in my case i need to start the timer to more than 10min)

Once everything is ready press the valid button. The controller board will switch on and off the oven automatically. It will go trough multiple step until it reach the IDLE step. L, v and ov parameter will be update at each step. Once the autotune reach the IDLE step you can press the valid button, it will go back to the Config menu with the updated parameters.



3.2 Reflow

This version of the controller only works with low temp solder paste SN42BI58. Other pastes may be implemented later.

First, apply solder paste on all the top mount footprint components. Place the top mount component in their right place. A stencil is a great help to place solder paste, but you can place it by hand with precaution.

Place the populated board in the oven, over or as close as possible to the temperature sensor

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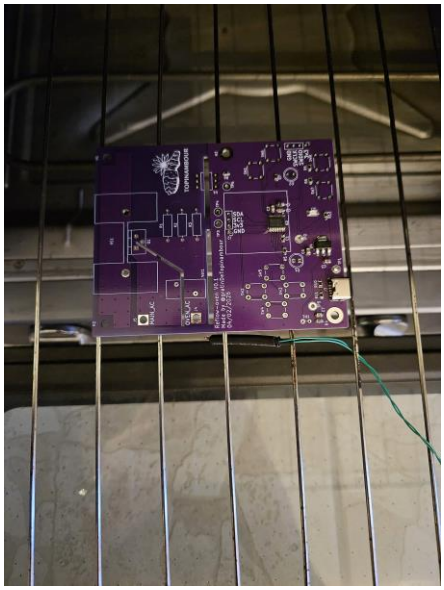
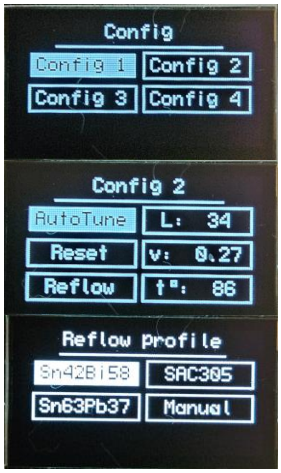


Then setup the temperature to 200°C and start the oven for more than 20min

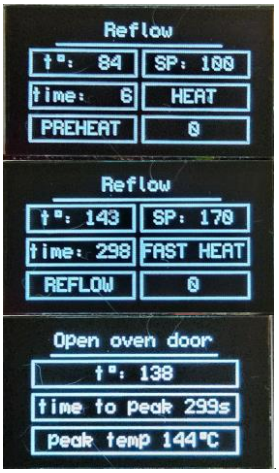
Start the controller board.



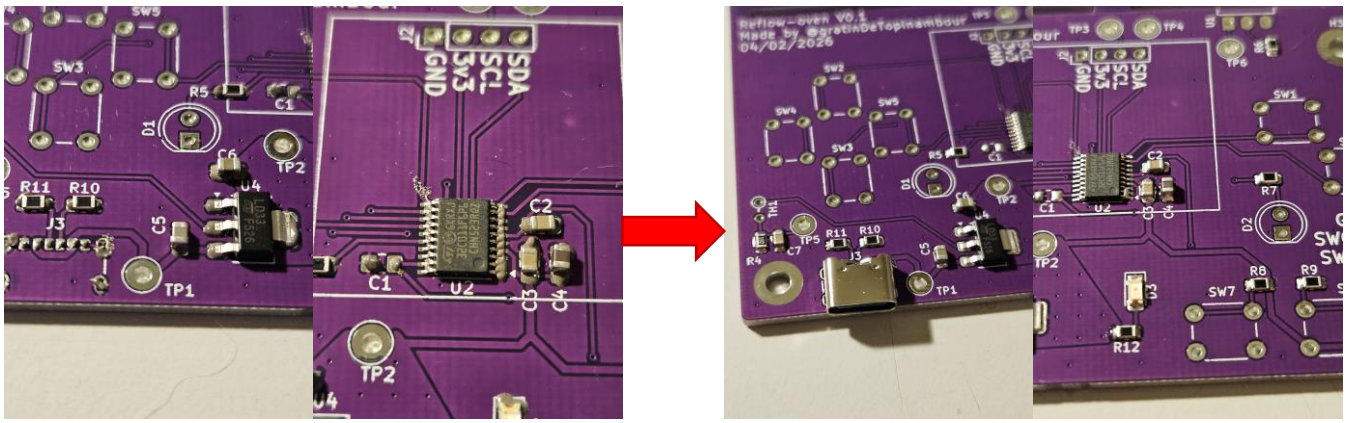
-chose a configuration and press valid

- check that L and v are not equal to zero then select reflow and press valid

- select reflow profile Sn42Bi58 other profile are not available



Once the board is correctly placed in the oven, close the door and press valid. the control board will turn on and off the oven to be has close has possible to the reflow profile. Once the reflow is done open the oven door to help the board cooling



Congratulations you've reflowed your first board! Some adjustments may need to be done manually. Make sure to check your board before using it.

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